

Ho Ko

AI/ML SOFTWARE ENGINEER

CONTACT

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SKILLS

Languages

Python C/C++ Swift Java
Kotlin JavaScript x86 Assembly
MATLAB

ML Lifecycle

Model Training Distillation
Quantization & Compression
Inference Engines
On-Device Deployment
Scalable Serving

On-Device & Web

WebGPU Three.js
Transformers.js MediaPipe
WebRTC Local LLMs (Gemma, LFM)

Domains

NLP Computer Vision Healthcare
Astronomy Weather Forecasting
Economics

EDUCATION

M.S. Computer Science

University of Southern California — AI & ML specialization
2021–2023 · GPA 4.0/4.0

M.S. Electrical & Computer Eng.

University of Waterloo — Machine Learning & AI
2020–2021 · GPA 3.9/4.0

B.S. Computer Science

National Central University, Taiwan
2016–2020 · GPA 3.8/4.0

AI/ML Software Engineer with cross-domain experience spanning astronomy, weather forecasting, economics, healthcare, NLP, and computer vision. Works across the full **model lifecycle** — training and distilling deep models, architecting highly efficient inference engines, and building scalable, high-throughput serving — and brings that intelligence **on-device** through WebGPU, Three.js, and local LLMs in privacy-preserving systems that have reached **1B+ users**.

EXPERIENCE

- **Machine Learning Engineer · Google** 2026–Present
 - Build **knowledge-grounded retrieval & ranking models** on Google Lens that power AI Mode in Google Search and AI Overviews at Search scale.
 - **Distill Gemini** into highly compressed models that hold near-equal quality while meeting the Lens stack's latency, scalability, and multimodal requirements.
- **Software Engineer · Google** 2023–2026
 - Core contributor to the **LiteRT LM inference engine**, running LLMs privately and offline across Windows, macOS, Linux, and Android — including NPU acceleration.
 - Enabled Gemma multimodality (vision + audio + text) and Gemini Nano on-device; helped ship **Chrome Built-in AI** and **Google AI Core**, reaching **1B+ users**.
 - Shipped the **Google AI Edge Gallery** app on Android and iOS.
- **Research Collaborator · Google DeepMind** 2024–2025
 - Designed **Sparse RAG (ICLR 2025)**, a sparse decoding paradigm that encodes retrieved docs in parallel and attends only to the most relevant caches via learned control tokens.
 - Cut long-range attention latency while **improving generation quality across 4 retrieval benchmarks**.
- **Research Collaborator · Google Health & Fitbit** 2024–2025
 - Built and evaluated ML models **detecting hypertension risk from raw wrist PPG** passively collected on a consumer smartwatch (medRxiv), enabling non-invasive screening at population scale.
- **Software Engineer · MyCareLinq** 2023
 - Built **NLP pipelines** and a **home-health-care knowledge graph** from unstructured health documentation.
 - Integrated LLMs powering caregiving assistants that surface contextual recommendations and care-plan summaries.
- **Software Engineer Intern · Google** 2022
 - Built a TFLite tensor extractor + metadata-writer with classification rules at **100% coverage of common GenAI task libraries** (LiteRT Task Library).
 - Integrated a **C++ code generator** compiling optimized inference pipelines for arbitrary TFLite files.
- **Research Assistant · Caltech (SURF)** 2018
 - Analyzed transient imaging from the Zwicky Transient Facility; **co-authored a Type IIb supernova study in ApJL (878:L5)** and simulated double-peak light-curve models.

SELECTED PUBLICATIONS

- *Accelerating Inference of Retrieval-Augmented Generation via Sparse Context Selection* — **ICLR 2025**, Google DeepMind.
- *Opportunistic Hypertension Detection from Wearable PPG* — **medRxiv 2025**, Google Health & Fitbit.
- *A Type IIb Supernova (ZTF18aalrxas)* — **ApJL 878:L5**, 2019.
- *Real-world WebAssembly Performance for ML* — **ACM IMC 2023**.

SELECTED PROJECTS

- **AI Mesh** — P2P network of on-device AI agents over WebRTC; each browser tab runs a local LLM + MCP knowledge base and gossips queries across the mesh.
- **The Match Maker** — Webcam multiplayer game running Gemma 3n fully client-side via WebGPU at ~15 tok/s with MediaPipe face mesh.
- **Tetris AI in x86 Assembly** — Self-learning heuristic AI in ~2,000 lines of hand-written x86, averaging 50,000 cleared lines.